

Design System & UX

polycarpic — personal budgeting & wealth management (virtual Single Family Office)

Owner: ux-designer agent **Status:** Draft — Plan phase, matured alongside ARCH **Version:** 0.6

Last updated: 2026-09-30

CONTENTS

- 1. Overview & Design Principles
- 2. Design System
- 3. Design Tokens
- 4. Components & Screen Styles
- 5. User Flows
- 6. Wireframes
- 7. Styled Screens
- 8. Accessibility
- 9. Open Questions
- 10. Appendix
 - 1. FAQs

1. Overview & Design Principles

What this product should feel like to use, and the principles that resolve design disputes.

- **Mobile-first when in doubt.** The primary v1 target is a self-hosted web dashboard used on desktop for deep review (statements, lot tables) and on mobile for quick checks (dashboard glance, confirming a draft transaction) — design the dense screens up from a mobile layout, not down from desktop.
- **Accessible by default (WCAG AA).** Non-negotiable per PRD §6. Financial gain/loss and status signals never rely on color alone — see [Financial semantics](#).
- **No decorative loading states.** Skeletons match the real layout; spinners are a last resort (see [Component inventory](#)).
- **Every number earns trust.** This is a general ledger, not a budgeting toy — a balance, a delta, or a pending-liability figure must always be traceable to its source (the journal entry, the lot, the reconciliation). The UI never shows a number it can't explain if clicked.
- **Immutability is visible, not just enforced.** A posted transaction reads as posted (badge, disabled edit affordance); a correction reads as a reversal + new entry, never as a silent edit (PRD §5).
- **Diverge from polycarpic's own tooling, deliberately.** The cairn tracker (our internal tool) and the product are not the same thing and shouldn't look like it — this system's identity is derived from **measured** screenshots and live style reads off Monarch Money and Origin (``useorigin.com``), not memory (Principal direction, 2026-09-28) — see [Inspiration: Monarch & Origin](#).
- **One brand hue, used honestly.** The Principal's neon-green brand color (`#2CFF05` , 2026-09-30 — "the project name relates to fruiting plants") measures 1.36:1 against white, far below AA — kept as the literal `--brand` value everywhere it clears AA (dark surfaces), with a darkened same-hue green for light-mode `--primary` , stated plainly rather than silently swapped for a different hue — see [Foundations](#).

2. Design System

The single source of truth for the visual language. The written spec lives alongside this doc; the machine-readable system lives in the two CSS files that `frontend-lead` consumes directly.

- `design-system-spec.md` § Inspiration — what was taken from Monarch Money and Origin, and why
- `design-system-spec.md` — the written spec (token rationale, financial semantics, component inventory, accessibility)
- `tokens.css` — design tokens as CSS custom properties (hand-authored palette, not a vendored shadcn preset)
- `screen.css` — component & screen styles built on the tokens

Figma stays the high-fidelity design tool for anything shipping pixel-exact (via the `ux-designer`'s Figma skills); `docs/DESIGN/` is the in-repo, code-level home that ships with the project and is the contract for implementation.

3. Design Tokens

Color, typography, spacing, radius, and shadow primitives — SvelteKit + shadcn-svelte (PRD §8). Edit values in [tokens.css](#); the swatches below preview the live variables. Full rationale in [design-system-spec.md](#).

3.1 Surfaces & Text

--background

--foreground

--card

--popover

--muted

--muted-foreground

3.2 Actions & States

--primary

--brand (dark surfaces only
– see Foundations)

--brand-2 (pass 5, same rule
– see Secondary accent)

--secondary

--accent

--destructive

3.3 Financial semantics (new — polycarpic-specific)

--positive

--warning

--destructive (reused for negative/danger)

See [design-system-spec.md § Financial semantics](#) — gain/loss, safe-harbor status tiers, and the "never color alone" rule.

3.4 Borders & Focus

--border

--input

--ring

3.5 Charts (sequential ramp — magnitude, not category)

--chart-1

--chart-2

--chart-3

--chart-4

--chart-5

3.6 Sidebar (inverted, pass 4; re-hued blue, pass 5)



`--sidebar`

`--sidebar-accent`

`--sidebar-ring`

Dark-on-light in light mode, light-on-dark in dark mode — see [design-system-spec.md § Sidebar](#) for the collapse-to-rail behavior and three proposed expand/collapse cues.

Type: `--font-sans` *Inter* — one grotesque sans for body and headings, hierarchy by weight, confirmed against both reference products' actual app UI (their serif accent is marketing-only, see *Inspiration*); `--font-mono` *Roboto Mono* — Origin's own measured choice, for uppercase eyebrow labels, account numbers, tickers, tabular figures. **Radius:** measured `--radius-sm|md|lg|xl` = 6/8/12/20px (cards 12px, buttons 8px — both products' actual app-level values, not either's marketing-only pill).

4. Components & Screen Styles

The component inventory and the styles that realize it, built on the tokens. Styles live in [screen.css](#); full states-per-component table lives in [design-system-spec.md § Component inventory](#).

Component	shadcn-svelte primitive	polycarpic usage
Net-worth hero	Chart (LayerChart, Area)	Big figure + signed delta on the figure + single-line area chart — direct Monarch pattern
Grouped account list	Table / custom	Type-grouped rows with sparkline + freshness — direct Monarch pattern
Segmented bar + legend	Custom, tokens-driven	Assets/Liabilities, allocation actual-vs-target — direct Monarch pattern
Badge	Badge	Entity/account/transaction status — see Financial semantics
Transaction table	Table	Ledger, import staging, exceptions report
Entity switcher	Tabs / Sidebar	Switch between Person/Trust/Business/Household
Transaction detail	Sheet	Drill into a posted/draft entry, its journal lines, source doc
Empty state	Card + plain text, no skeleton	Dashboard panels not yet shipped (tax at 1.0, allocation at M4) — see styled-screens/dashboard.html
Header (pass 5)	—, custom full-width bar	Brand lockup left, entity selector right; page title/actions moved to the content column — see design-system-spec.md § Header
Avatar cluster (pass 5, new)	—, custom	Overlapping initials in the entity selector, max 3 + overflow
Public hero (pass 4, corrected)	—, custom	Home — pre-auth landing page, product statement + CTAs, not a signed-in screen

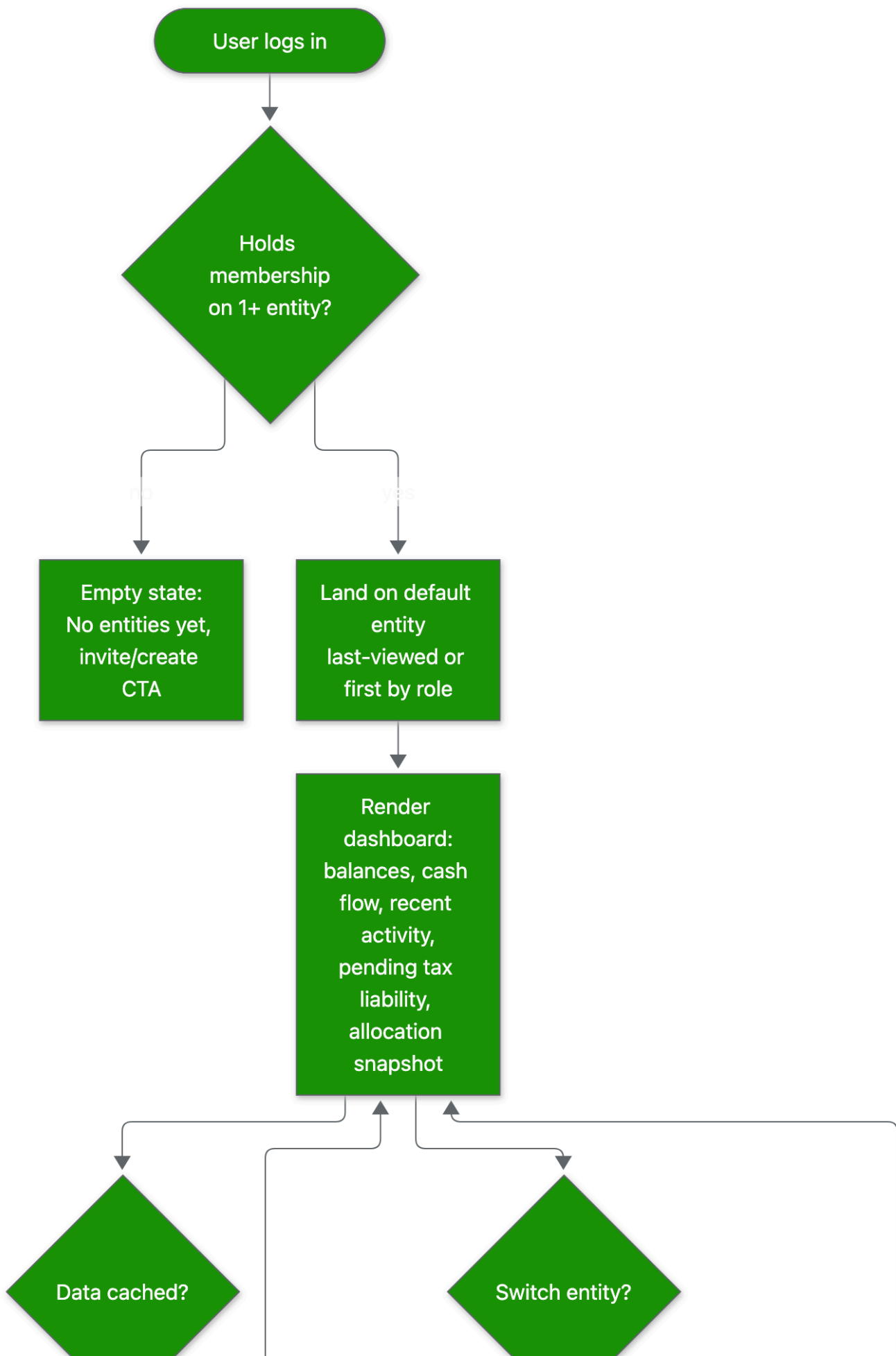
Component	shadcn-svelte primitive	polycarpic usage
Split screen login <i>(pass 4, refined pass 5)</i>	—, custom	Login — full field list, centered form column, always-branded visual pane

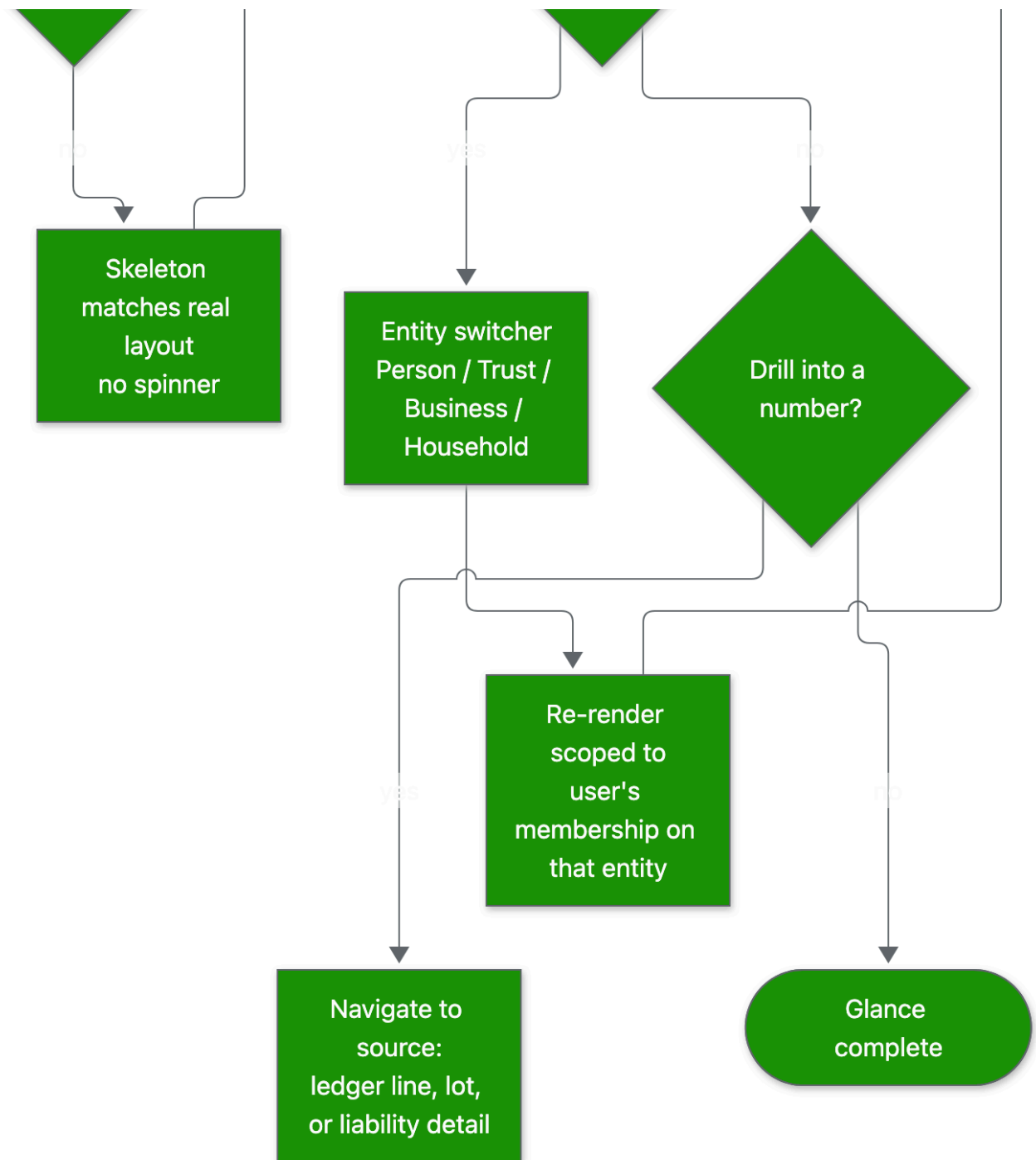
5. User Flows

How users move through the product to accomplish the highest-traffic stories (PRD §4). **Pass 5 fix (2026-09-30):** these previously linked to `flows/*.md`, which the browser serves as raw Markdown — the fenced `mermaid` blocks never ran through the page's own Mermaid loader, so nothing rendered (not a syntax error, as first suspected). Each flow is now embedded inline below; the `.md` files stay as the diffable source of truth in flows/.

5.1 Dashboard glance

Story (PRD §4.1): per-entity financial health at a glance. Source: flows/dashboard-glance-flow.md.

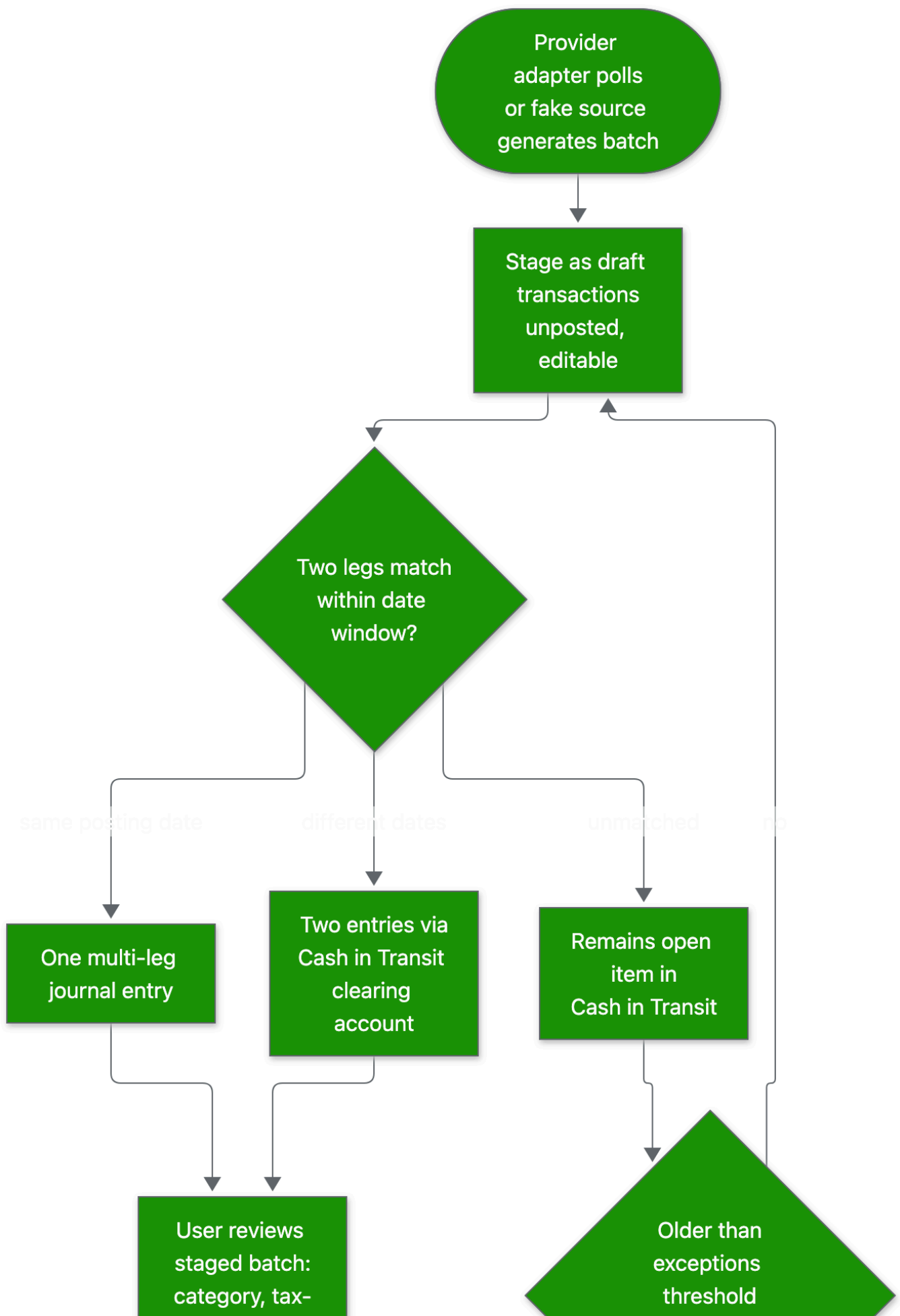


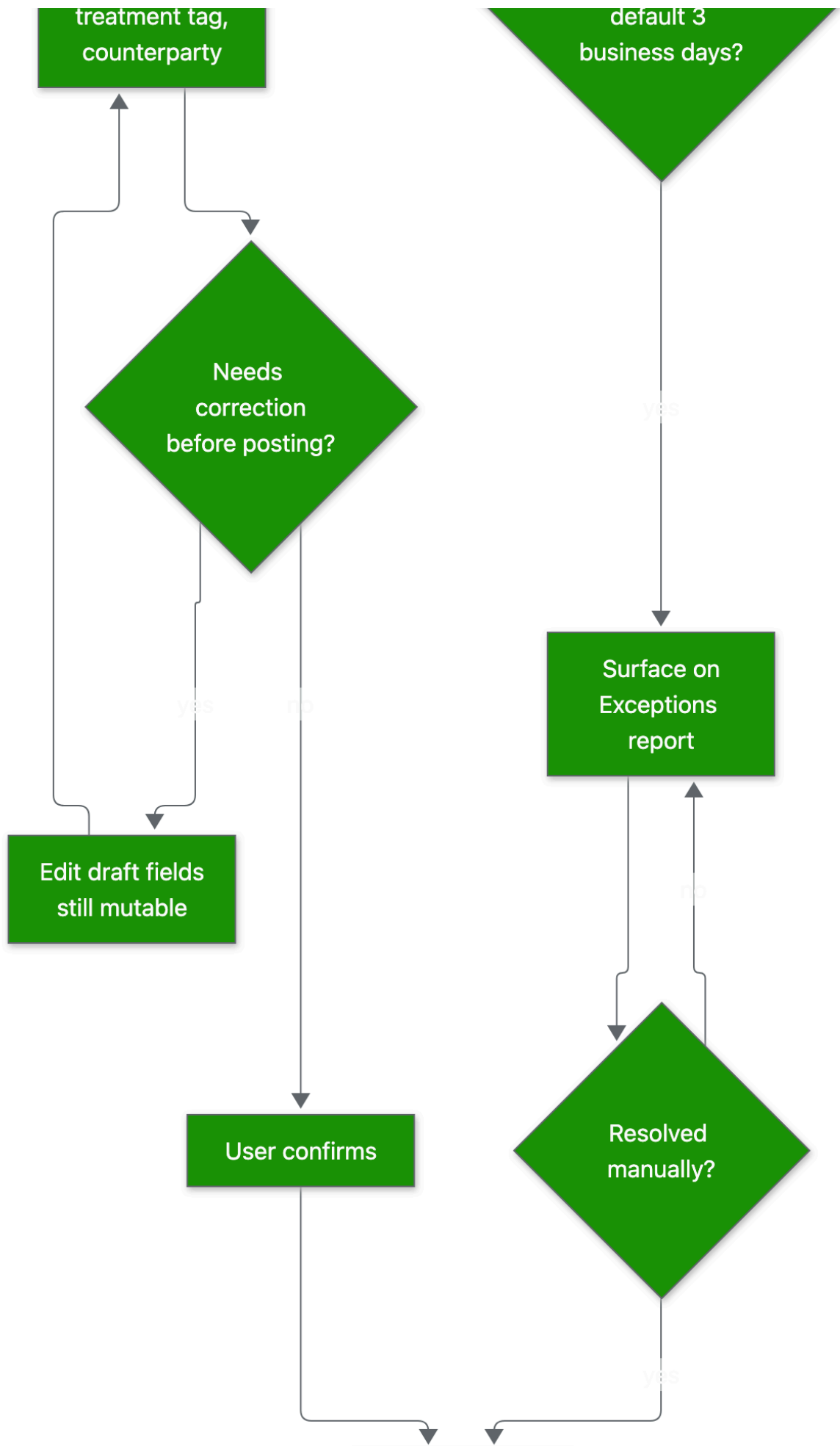


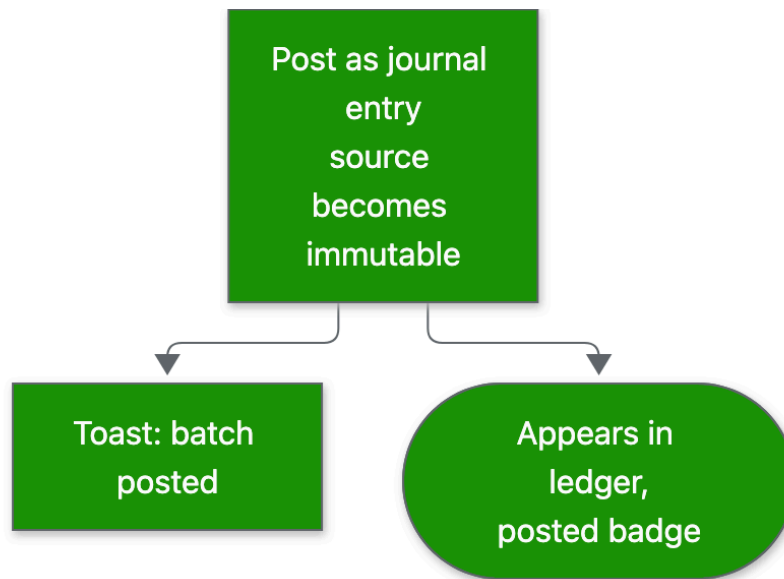
- Every number on this screen must be clickable through to its source (Overview principle: "every number earns trust").
- Filtered strictly by membership (owner/manager/viewer) — no transitive access via parent entity (PRD §5 permission model).

5.2 Import & confirm transactions

Story (PRD §4.2): *staged drafts* → *reviewed* → *posted*. Source: *flows/import-confirm-flow.md*.



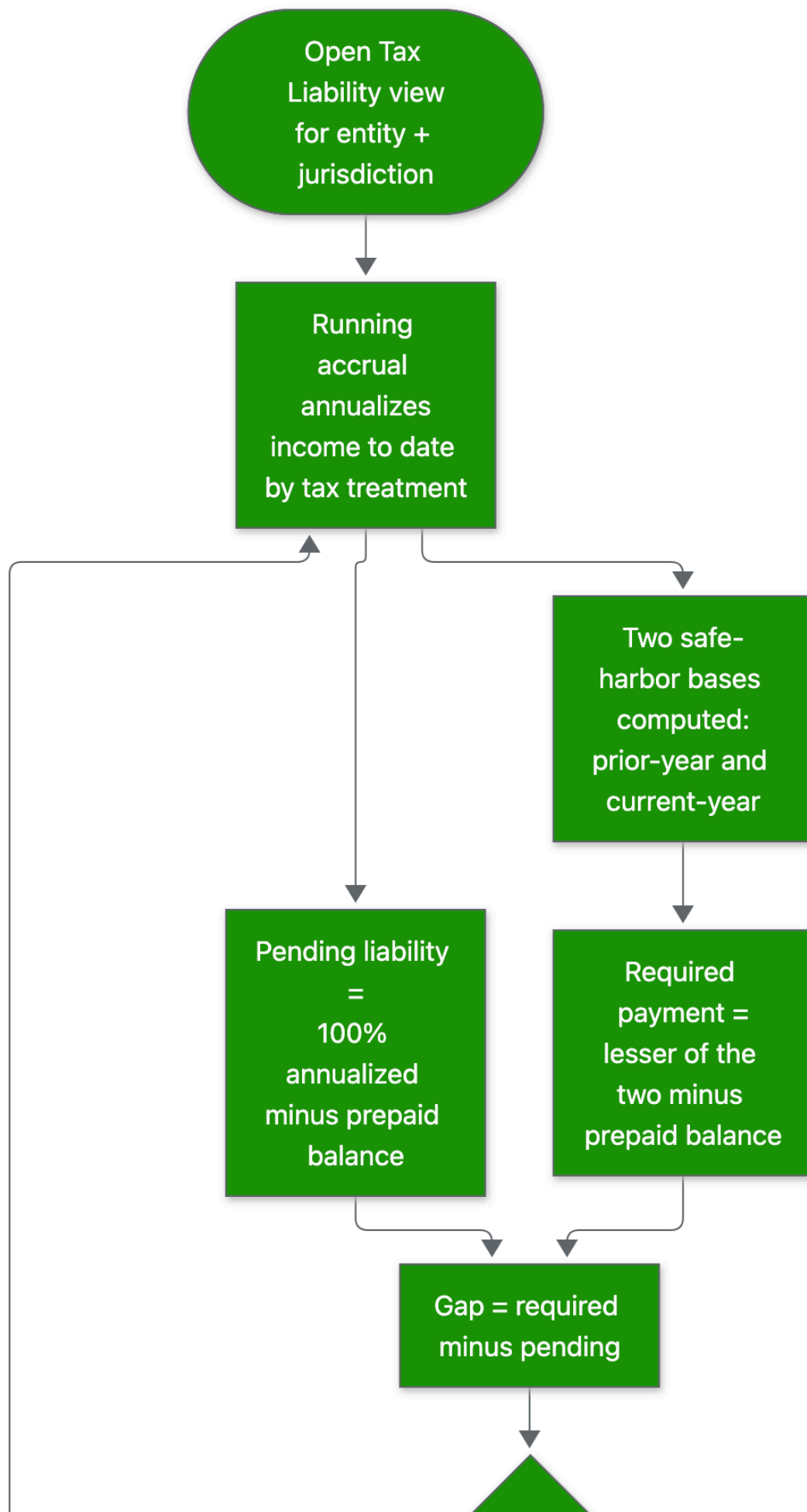


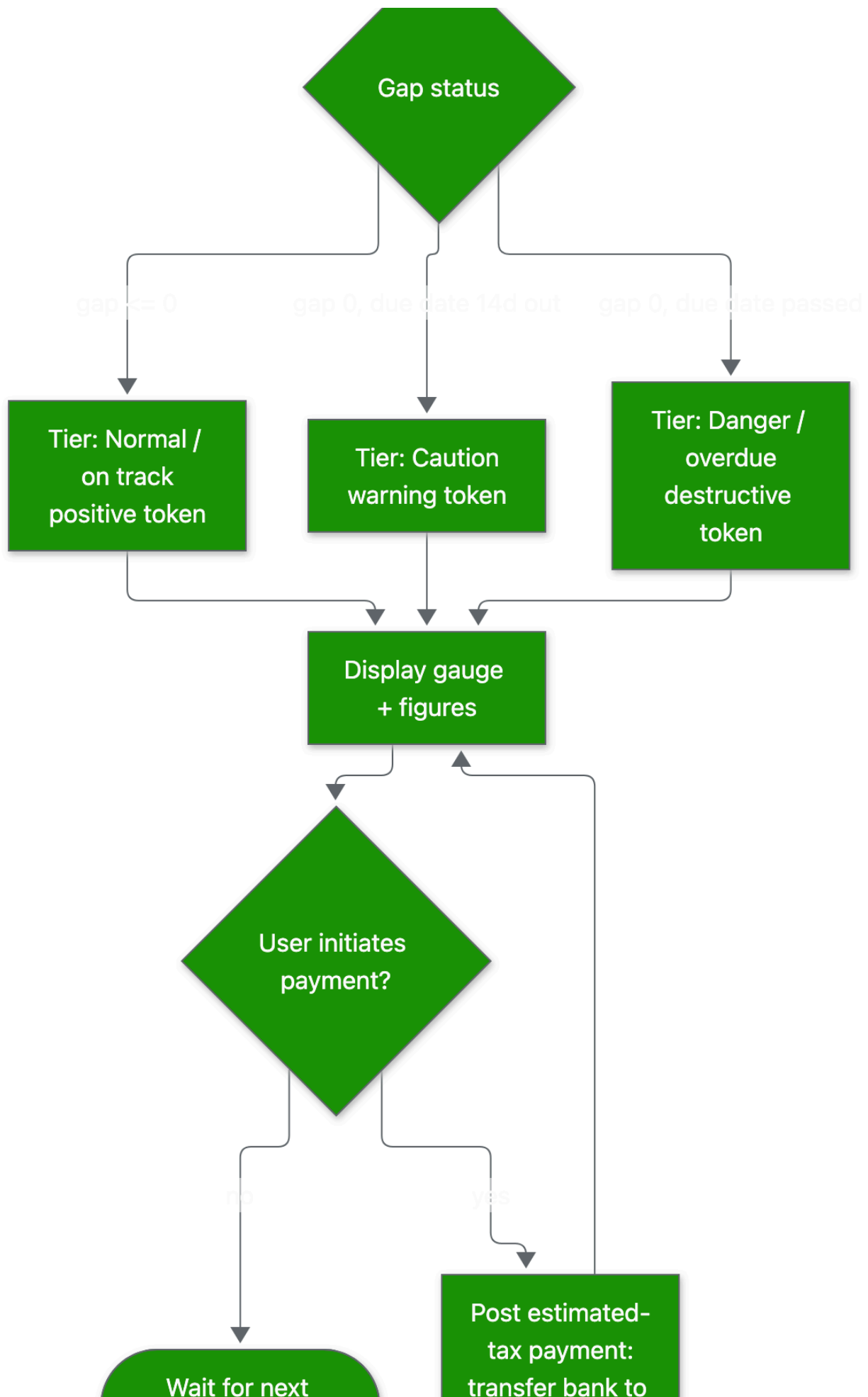


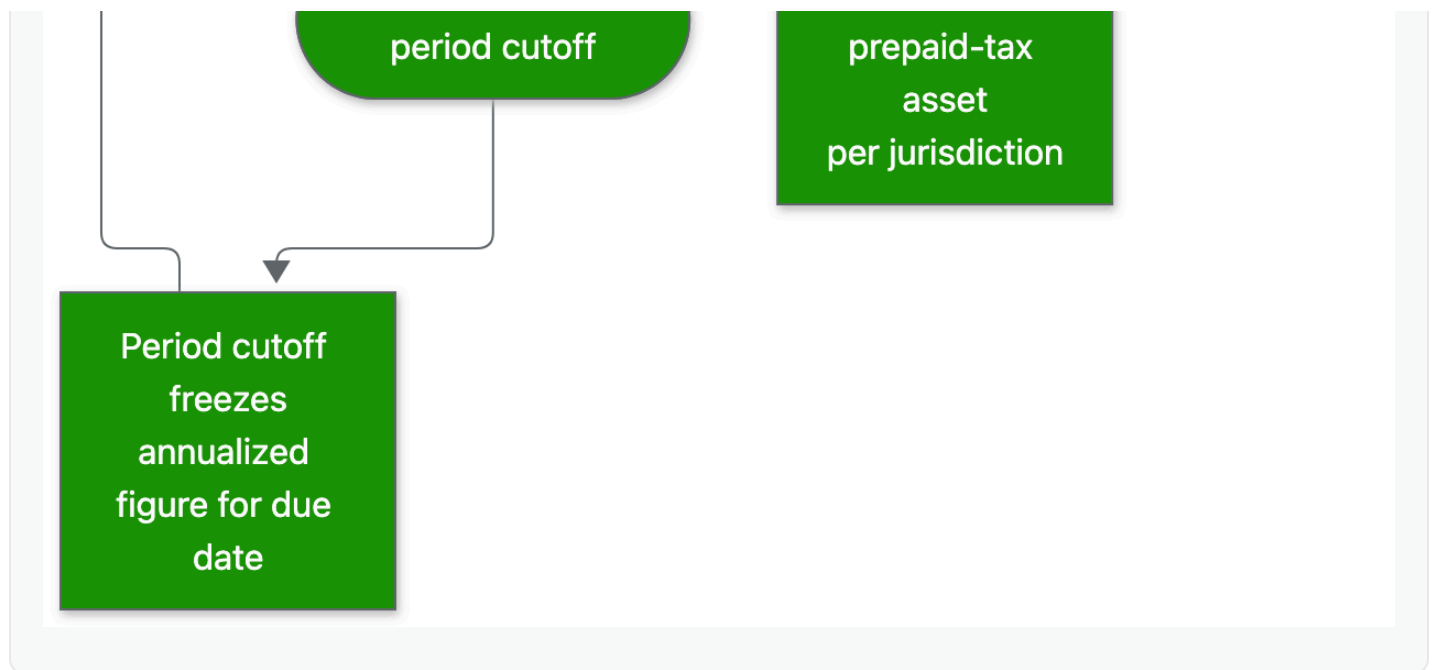
- Once posted, a transaction is immutable — the UI never offers an "edit" affordance on a posted row, only "reverse".
- Actor attribution (user / auto-post rule / system job) is shown on every draft and posted row, never inferred (PRD §5).

5.3 Tax liability & safe-harbor

Story (PRD §4.4): *pending liability, required payment, gap, pay.* Source: *flows/tax-liability-flow.md*.







- The three-tier status (normal/caution/danger) reuses the product-wide status vocabulary — see [design-system-spec.md § Financial semantics](#).
- Jurisdiction due dates, multipliers, and cumulative percentages are data-driven, so this view must render correctly for an arbitrary number of jurisdictions (PRD §5).

5.4 Securities lots & balance sheet reconciliation

Stories (PRD §4.5–4.7): *GL vs. custodial reconciliation, allocation vs. target.* Source: [flows/securities-lots-flow.md](#).

Open Securities
view
for entity



Lot table per
custodial
account:
book value,
market value,
cost basis



Daily EOD price
feed
updates market
value

Balance sheet
basis



GL



Custodial

Balance sheet

Balance sheet
by GL account

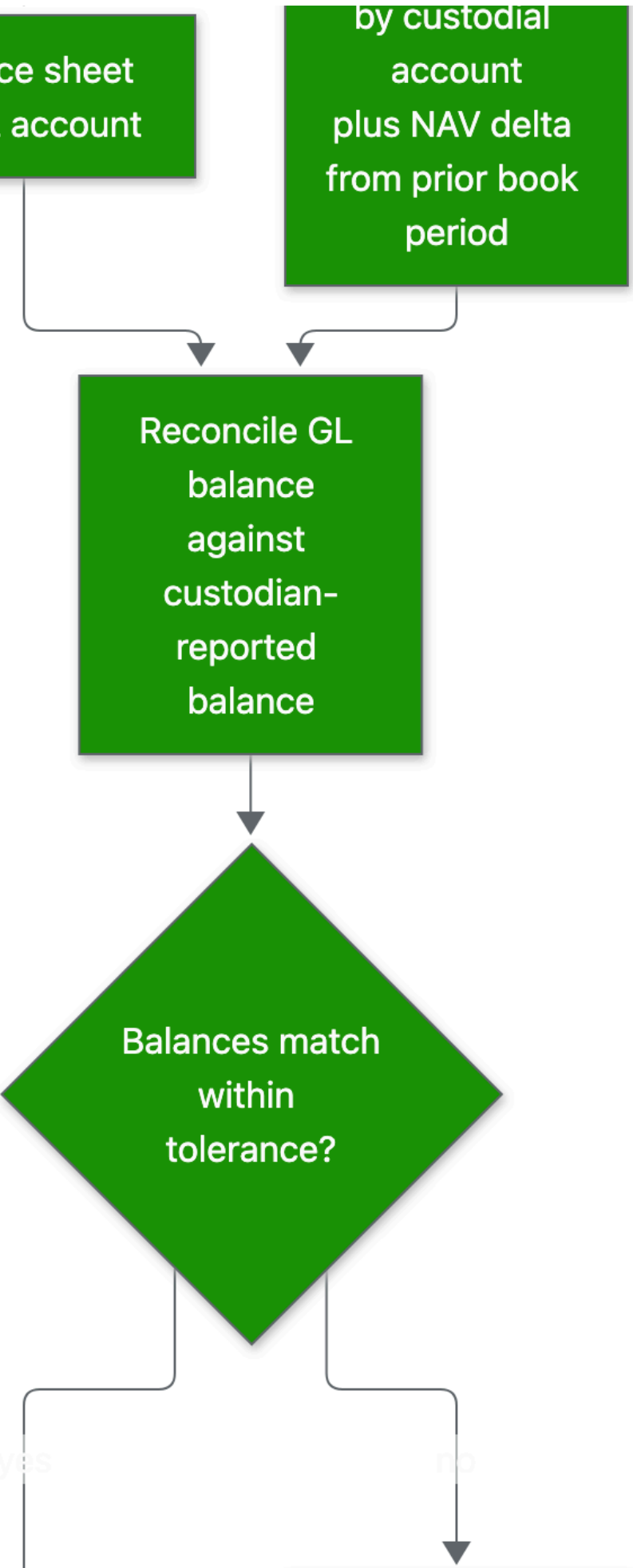
by custodial
account
plus NAV delta
from prior book
period

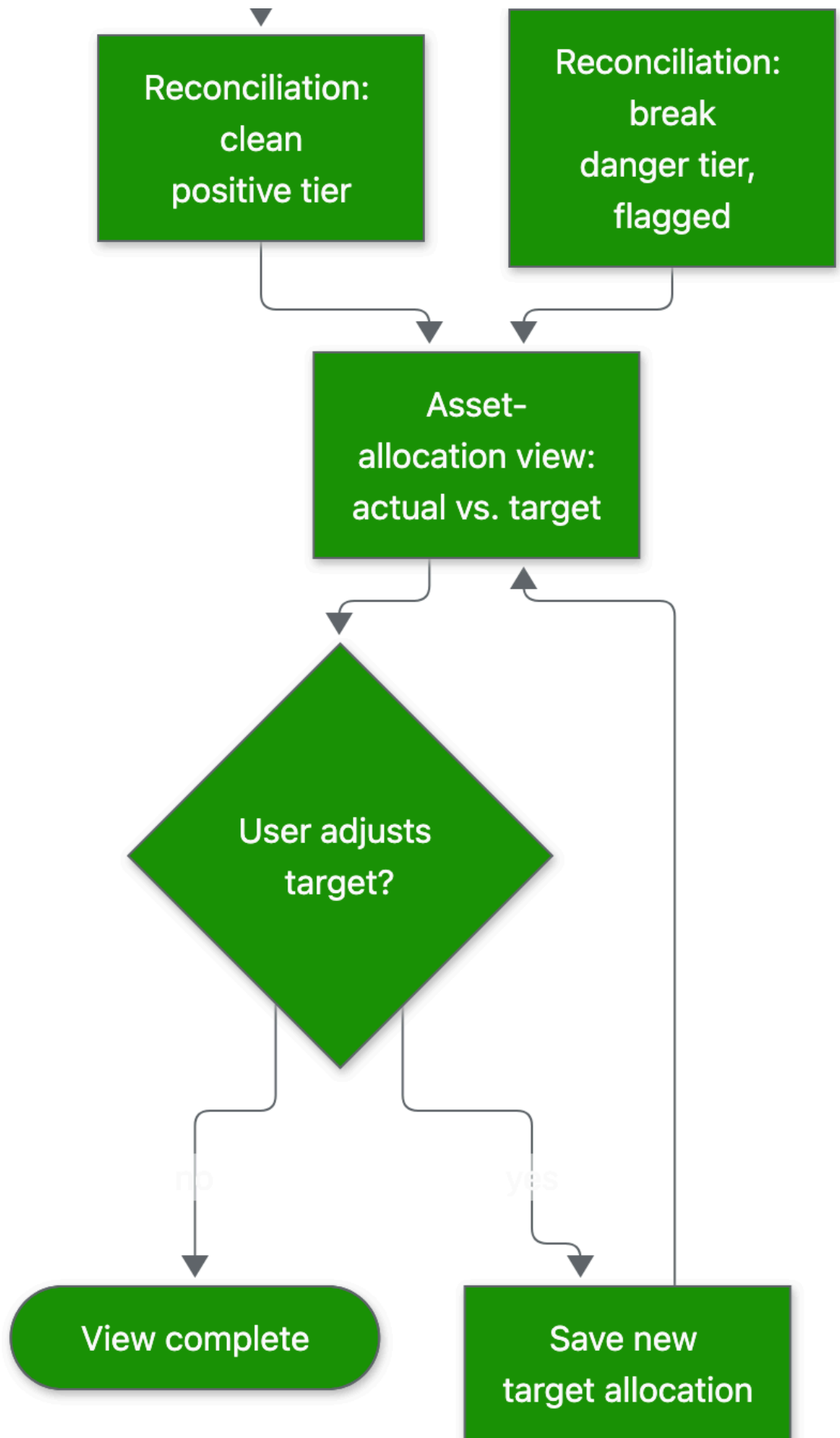
Reconcile GL
balance
against
custodian-
reported
balance

Balances match
within
tolerance?

yes

no

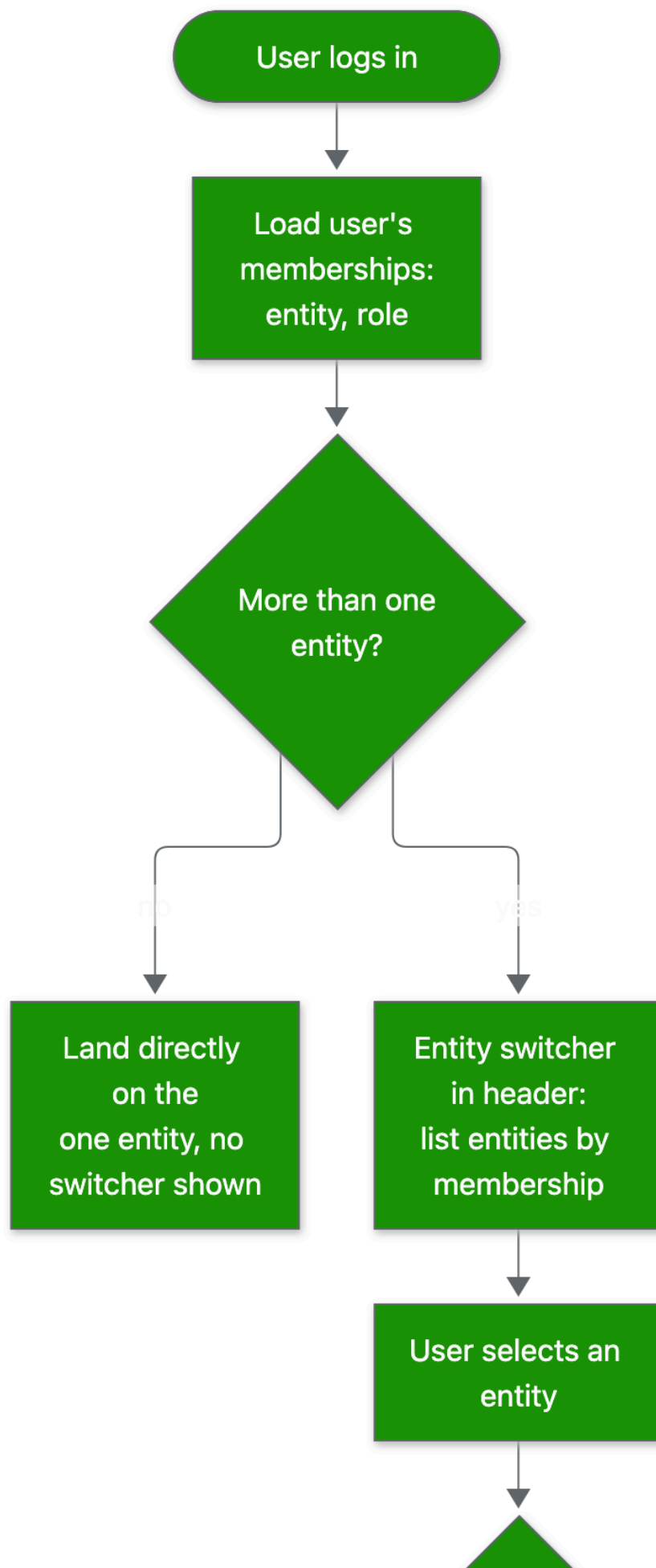


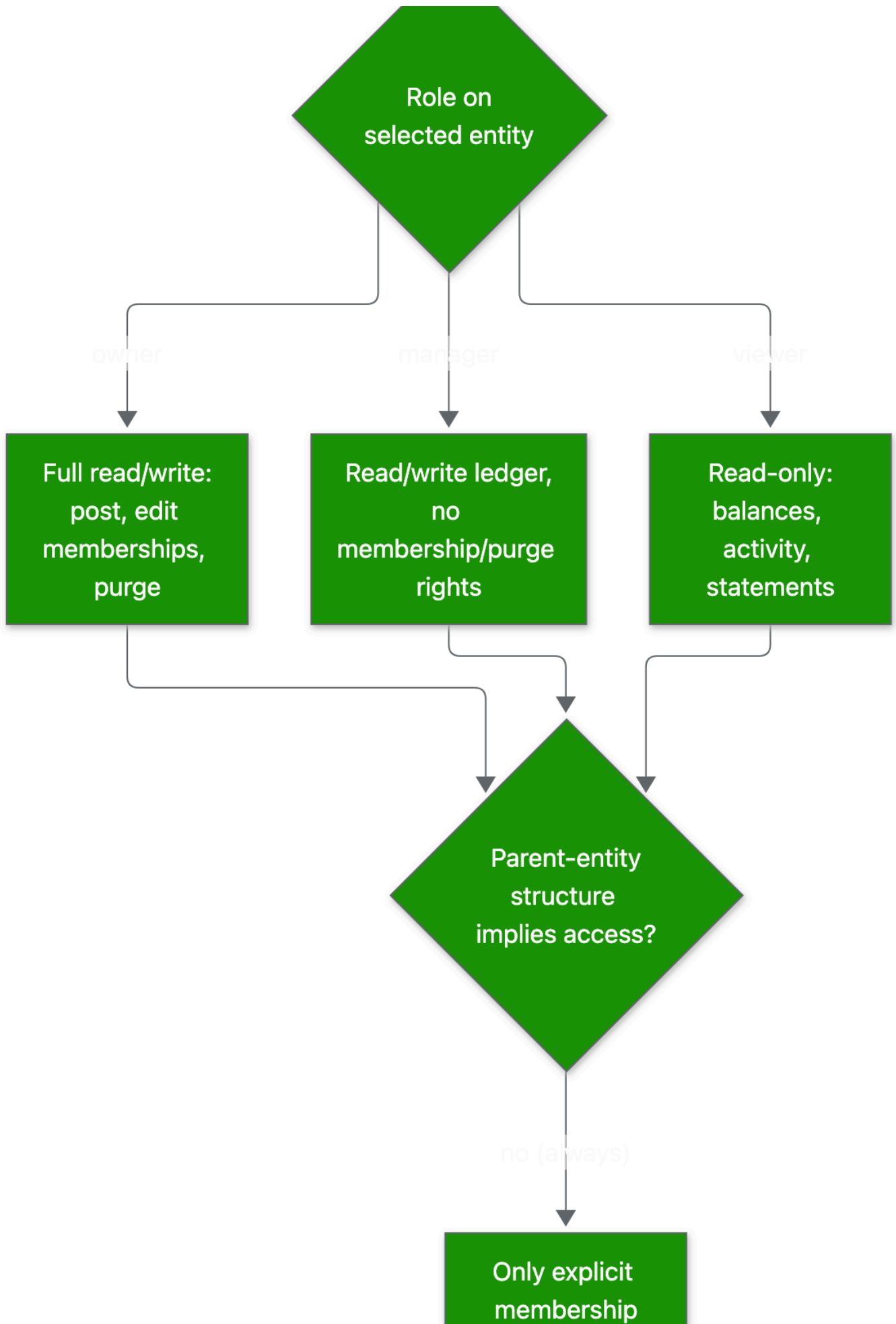


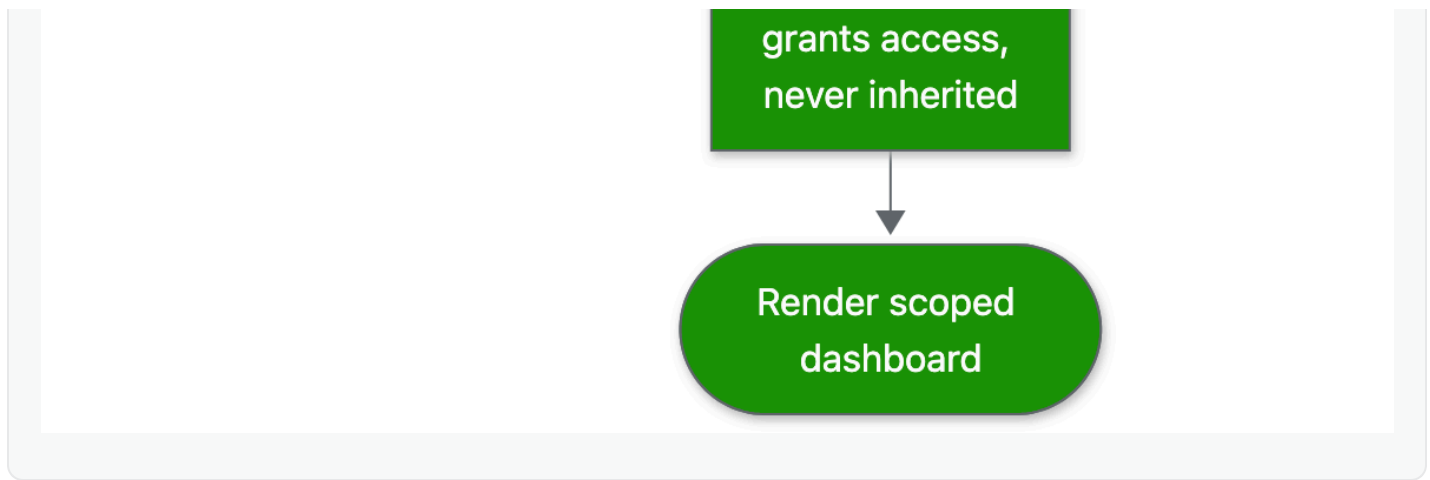
- GL-vs-custodial is a `Tabs` toggle on one screen, not two separate pages.
- A reconciliation break uses the danger tier (`--destructive`) and must surface on the Dashboard glance too.

5.5 Entity switch & membership access

Story (PRD §4.8): *owner/manager/viewer boundary. Source: [flows/entity-membership-flow.md](#).*







- A Household's membership never transitively grants access to a member Person entity, or vice versa (PRD §5, hard requirement).
- A viewer never sees a write affordance rendered-then-disabled — write controls are omitted entirely, not shown-and-greyed.

6. Wireframes

Low-fidelity layout of each key screen — structure before style. Assets in [wireframes/](#).

- [Dashboard](#) — per-entity financial health
- [Ledger / Transactions](#) — import staging + confirm
- [Accounts](#) — GL vs. custodial balance sheet
- [Entities / Memberships](#)
- [Securities / Lots](#) — lot table + allocation view
- [Import](#) — provider link + batch review
- [Home](#) (*pass 4, new*) — entity picker, skipped for single-entity users
- [Login](#) (*pass 4, new*) — two-pane, Origin-inspired
- [Sidebar expand/collapse cues](#) (*pass 4, decided pass 5*) — option A (chevron) chosen, B/C kept for the record

7. Styled Screens

High-fidelity screens that apply the design system to the wireframes, consuming `tokens.css` + `screen.css` — a visual regression reference and a head-start for `frontend-lead`. Prioritized by traffic (PRD §4 stories 1 & 2); remaining screens promote from wireframe fidelity as the design system stabilizes.

- Dashboard
- Ledger / Transactions
- Home (pass 4, new)
- Login (pass 4, new)

8. Accessibility

Accessibility is a design responsibility, not a QA afterthought. Full table in [design-system-spec.md § Accessibility](#).

Concern	Standard / approach
Color contrast	WCAG AA (4.5:1 text, 3:1 large/UI) — every token pair re-measured for the pass-4 neon-green rebrand, both modes; <code>--brand</code> deliberately fails against light surfaces (1.36:1, by design — never used there, see Foundations); no CVD simulation run yet (open item, now higher-priority given how central the brand hue is)
Color-alone signaling	Never — every gain/loss/status pairs color with a sign/icon/label
Keyboard	All interactive elements reachable & operable (shadcn default)
Screen readers	Semantic markup; labelled controls; delta sign readable from text content
Motion	Respect <code>prefers-reduced-motion</code> (policy not yet written)

9. Open Questions

Unresolved design decisions and assumptions that could be wrong. Batched to team-lead for relay to the Principal — not blocking.

- **Resolved 2026-09-30 (pass 5):** sidebar expand/collapse cue — option A (chevron) chosen; sidebar now full-height; header restructured to brand lockup + entity selector with an avatar cluster; login form pane centered — design-system-spec.md § Header, § Sidebar, § Login.
- **Avatar cluster overflow (pass 5):** the "+N" fold for 4th+ members is specified but not yet shown live in either reference screen (both currently have ≤ 2 members) — worth a quick visual check once a real multi-member entity exists.
- **Login's visual pane (pass 4):** built as a brand gradient, not a photo — no photography asset exists for this product. A real photo/illustration asset (in the style of the captured `dribbble` reference) would likely read stronger; flagging as a design-asset gap, not a layout question.
- No CVD (color-vision-deficiency) simulation run on the new palette yet — now higher priority given how central the brand hue is (recommended before lock, design-system-spec.md § Accessibility).
- The Accounts screen's full grouped-account-list + segmented-bar treatment was only built on the Dashboard styled screen (the highest-traffic comparison point) — Accounts itself is still at wireframe fidelity; promote it the same way as a fast-follow if the gap matters before Implement.
- Native iOS/macOS token export format (kickoff §2.10) — not yet decided, non-blocking for web.
- `prototype/` throwaway UX prototype (kickoff §2.9) not yet built — confirm ownership/timing with team-lead.

10. Appendix

- PRD user stories & functional requirements: `docs/PRD/index.html` §4–5
- Kickoff record, frontend/design system decisions: `docs/project_kickoff.md` §2.9

10.1 FAQs

Q: How do I add review comments to this design doc?

A: Hand-edit `docs/DESIGN/comments.md` with `## §<section-id>` blocks, or run `/serve-docs DESIGN` and use the inline `+ Comment` widget. Then run `/refine-doc DESIGN`. See `process/WORKFLOW.md` → Doc review loop.

Q: Why is the design system in `docs/DESIGN/` and not in the app's source?

A: It's the design *deliverable* — the contract. During Implement, `frontend-lead` imports or derives the app's real stylesheets from `tokens.css` / `screen.css`.

Q: Where did `variants.css` go?

A: It was generated infrastructure for the `cairn` tracker tool's own theme switcher, never part of the product — removed from this directory (POLY-78 relocates its generator target the same way). See `design-system-spec.md` § Provenance & shared tooling.